

1.1 Dataset Selection

The dataset used in this project is a demand forecasting dataset that represents retail sales data across multiple stores, products, and regions over time. This type of data is particularly interesting because it reflects real-world business challenges related to predicting demand, managing inventory, and optimizing operations. Understanding how demand changes over time and across different factors is essential for making informed business decisions.

I chose this dataset because it allows for meaningful analysis of trends, comparisons between stores and products, and identification of patterns in demand. It also includes a variety of variables such as demand, price, and units sold, which provide opportunities to create useful metrics and visualizations. This makes it suitable for building a comprehensive and interactive dashboard.

The dataset was obtained from Kaggle and is designed to simulate real-world retail demand forecasting scenarios. It includes both historical sales data and additional factors such as category, region, and pricing, which help in analyzing how different elements influence demand.

1.2 Variable Identification

The dataset contains a mix of categorical, numerical, and temporal variables that support comprehensive demand analysis across different dimensions such as time, location, and product characteristics.

◆ Categorical Variables

Categorical variables represent qualitative data used to group and compare different segments:

- **Store ID** – Identifies different store locations
- **Product ID** – Represents individual products
- **Category** – Classifies products into groups
- **Region** – Indicates the geographical location
- **Weather Condition** – Represents external environmental conditions

◆ Numerical Variables

Numerical variables represent measurable quantities used for analysis and calculations:

- **Demand** – Total number of units required
- **Units Sold** – Quantity of products sold
- **Units Ordered** – Quantity of products ordered
- **Price** – Price per unit of the product
- **Discount** – Discount applied to the product
- **Inventory Level** – Available stock level
- **Competitor Pricing** – Price offered by competitors

◆ Temporal Variable

- **Date** – Represents the time of transaction and is used for trend and time-series analysis

Variable Description Table

Variable Name	Type	Business Meaning
Date	Temporal	Indicates when the transaction occurred
Store ID	Categorical	Identifies store location
Product ID	Categorical	Identifies specific product
Category	Categorical	Groups products into categories
Region	Categorical	Represents geographical area
Weather Condition	Categorical	External factor affecting demand
Demand	Numerical	Number of units required (not monetary)
Units Sold	Numerical	Number of units sold
Units Ordered	Numerical	Number of units ordered
Price	Numerical	Cost per unit of product
Discount	Numerical	Reduction applied to price
Inventory Level	Numerical	Available stock quantity
Competitor Pricing	Numerical	Price offered by competitors

1.3 Expected Insights & DAX Measures

◆ Expected Insights

Before developing the dashboard, several key insights are expected from the dataset:

1. **Demand trends over time**

The dataset is expected to reveal how demand fluctuates across different time periods, helping identify seasonal patterns or peak demand periods.

2. **Store performance comparison**

Differences in demand across stores will help identify high-performing and low-performing locations.

3. **Product-level demand concentration**

A small number of products are expected to contribute significantly to total demand, indicating key revenue-driving items.

4. **Regional demand distribution**

Demand is expected to vary across regions, highlighting geographic differences in customer behavior.

5. **Impact of pricing and external factors**

Variables such as price, discount, weather conditions, and competitor pricing are expected to influence demand patterns.

◆ DAX Measures

To support the analysis, the following DAX measures were created:

1. Total Demand

Total Demand = SUM(demand_forecasting[Demand])

2. Total Sales

Total Sales = SUM(demand_forecasting[Units Sold])

3. Average Price

Avg Price = AVERAGE(demand_forecasting[Price])

4. Total Orders

Total Orders = SUM(demand_forecasting[Units Ordered])

◆ Importance of DAX Measures (with Citations)

DAX measures are essential for transforming raw transactional data into meaningful business metrics. (Sharda et al., 2020) Aggregated measures such as totals and averages help summarize large datasets and enable comparisons across time, products, and locations.

For example, total demand and total sales provide an overview of performance, while average price helps in understanding pricing trends. These metrics are crucial for decision-making in areas such as inventory management, pricing strategy, and operational planning.

According to Ramesh Sharda et al. (2020), business intelligence systems rely on aggregated metrics to convert raw data into actionable insights that support strategic and operational decisions. Similarly, measures such as averages and totals are widely used in data analytics to identify patterns, trends, and anomalies in business data. (Provost & Fawcett, 2013)

2. Dashboard Design & Construction

2.1 KPI Cards

The dashboard includes four KPI cards at the top to provide a quick overview of key business metrics: Total Orders, Total Sales, Average Price, and Total Demand. These KPIs were selected because they represent core aspects of business performance, including demand, revenue, and pricing strategy. Conditional formatting was applied to these KPI cards to visually highlight performance differences, allowing users to quickly interpret whether values are relatively high or low.

2.2 Dashboard Layout

A structured 3×2 grid layout was used to organize the dashboard. KPI cards are placed at the top for immediate visibility, followed by charts arranged below based on analytical importance. The line chart showing demand trends over time is positioned prominently to highlight temporal patterns. Supporting visuals such as store comparison, product-level analysis, and regional distribution are arranged logically to enable efficient data exploration.

2.3 Background and Theme

A light and neutral background was used to maintain readability and minimize visual clutter. A consistent grey color palette was applied across all visuals to create a cohesive and professional appearance. Accent colors such as green and red were used selectively in conditional formatting to represent performance differences without overwhelming the dashboard.

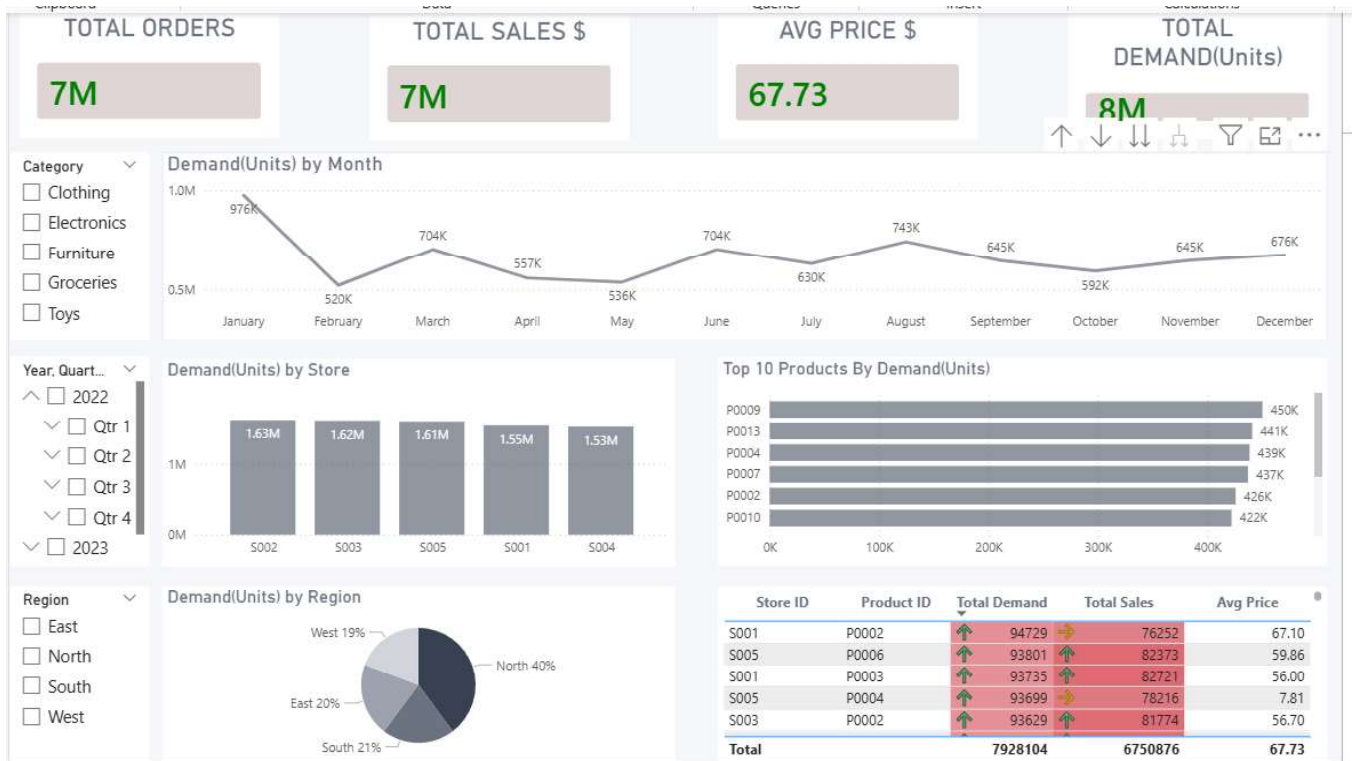
2.4 Cross-Filtering and Interactivity

Cross-filtering was implemented across all visuals in the dashboard to enhance interactivity and improve data exploration. All visual interactions were configured to use filtering rather than highlighting. This ensures that when a user selects an element, such as a region, product category, or store, all other visuals on the dashboard update dynamically to reflect the selected data.

For example, selecting a specific region filters the line chart, store comparison, and product-level analysis to display only data related to that region. Similarly, slicers allow users to filter the dashboard by category or time period, enabling more focused analysis. This interactive functionality helps users quickly identify patterns, relationships, and trends within the data.

2.5 Dashboard Levels

The dashboard follows a layered analytical structure. Level 1 consists of KPI cards that provide a high-level summary of performance. Level 2 includes aggregated visuals such as line charts and bar charts for trend and comparison analysis. Level 3 consists of the detailed table, which provides granular insights into product and store-level performance. This structured approach supports both quick decision-making and detailed analysis.



3.1 Business value and Extracted Insights

Dashboard Pitch

The demand forecasting dashboard is designed to provide a clear and interactive view of retail performance across multiple dimensions, including time, products, stores, and regions. The primary purpose of this dashboard is to transform raw transactional data into meaningful insights that support data-driven decision-making.

This dashboard helps address key business challenges such as demand fluctuations, inventory planning, and performance differences across locations and products. By visualizing demand trends over time, decision-makers can identify peak periods and plan inventory accordingly to avoid stockouts or overstocking. The comparison of store performance enables the identification of high-performing locations as well as stores that may require operational improvements. Additionally, product-level analysis highlights which products consistently drive demand, allowing businesses to prioritize inventory and optimize product strategies.

The target audience for this dashboard includes business managers, operations teams, and data analysts who are responsible for monitoring performance and making strategic decisions. The dashboard is designed to be user-friendly and interactive, allowing users to filter data by category, region, and time period. This interactivity enables users to explore specific scenarios

and gain deeper insights without requiring advanced technical skills.

One of the key strengths of this dashboard is its ability to integrate multiple perspectives into a single, cohesive view. Users can analyze trends, compare performance across different dimensions, and drill down into detailed data using the table. This reduces the time required to analyze data and improves the efficiency of decision-making processes.

Overall, this dashboard provides valuable insights that can help improve demand forecasting accuracy, optimize inventory distribution, and enhance overall business performance. By enabling quick access to key metrics and supporting interactive analysis, it allows organizations to respond more effectively to changing market conditions and customer demand patterns.

(Davenport & Harris, 2007)

Insight 1: Demand shows sharp early decline followed by mid-year recovery and stabilization

Source: Line chart showing *Demand (Units) by Month*

Insight:

Demand begins at a high of approximately 976K units in January, then drops significantly to 520K units in February, representing a decline of about 47%. It then recovers to 704K units in March, dips again to around 536K–557K units in April and May, and rises back to 704K units in June. The demand reaches a secondary peak of

743K units in August before stabilizing between 592K and 676K units from September to December.

Importance:

This pattern indicates strong volatility at the beginning of the year, followed by a more stable demand trend in later months. Understanding these fluctuations is critical for accurate demand forecasting and efficient inventory management. (Kimball & Ross, 2013)

Action:

Increase inventory and operational capacity before January and mid-year (June– August) peaks, and implement strategies to manage the sharp decline after peak periods.

Insight 2: Store demand is relatively balanced, with slight dominance by top-performing stores

Source: Bar chart showing *Demand (Units) by Store*

Insight 2:

Store S002 has the highest demand at approximately 1.63M units, followed closely by S003 (1.62M units) and S005 (1.61M units). The lowest-performing store, S004, still records a strong demand of around 1.53M units. The difference between the highest and lowest store is only about 100K units, which is roughly a 6% variation.

Importance:

This indicates that demand is evenly distributed across stores, suggesting consistent performance and stable operations across locations.

Action:

Maintain balanced inventory distribution across stores while slightly prioritizing higher-performing stores such as S002 and S003.

Insight 3: Demand is moderately concentrated among top products, with limited variation between them

Source: Column chart showing *Top 10 Products by Demand (Units)*

Insight:

Product P0009 has the highest demand at approximately 450K units, followed by P0013 (441K units) and P0004 (439K units). The lowest among the top 10, P0008, records around 397K units. The difference between the highest and lowest product is about 53K units, or roughly a 12% variation.

Importance:

This indicates that demand is evenly spread among top products, reducing reliance on a single product and improving business stability.

Action:

Ensure consistent inventory levels across all top products while slightly prioritizing higher-demand products like P0009.

Insight 4: Demand is highest in the North region, with relatively balanced distribution across other regions

Source: Pie chart showing *Demand (Units) by Region*

Insight:

The North region contributes the highest demand at approximately 3.2M units (40%), while South contributes around 1.6M units (21%), East about 1.6M units (20%), and West approximately 1.5M units (19%). The difference between the highest and lowest regions is about 1.7M units.

Importance:

This shows that the North region is the primary demand driver, while other regions contribute almost equally

Action:

Allocate more resources and inventory to the North region, while maintaining balanced strategies in other regions.

Insight 5: Price variations show limited direct impact on demand within similar product ranges

Source: Table showing *Product-level Demand, Sales, and Average Price*

Insight:

Products with lower prices do not always generate higher demand. For example, P0003 priced at \$56.00 shows high demand (93,735 units), while P0004 priced at \$100.21 has lower demand (70,725 units). Meanwhile, mid-priced products such as P0002 (\$67.10) and P0005 (\$59.28) still maintain strong demand (94,729 and 91,445 units respectively).

Importance:

This indicates that demand is influenced by factors beyond price, such as product type, customer preference, or promotions.

Action:

Use a combination of pricing strategies, promotions, and product positioning instead of relying solely on price adjustments to drive demand.

References

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(citations : Sharda et al., 2020; Provost & Fawcett, 2013; Kimball & Ross, 2013; Davenport & Harris, 2007)